

Project Design Phase

Problem – Solution Fit

Date	22 September 2026
Project Name	Online Complaint Registration and Management System

Problem – Solution Fit

1. CUSTOMER SEGMENT(S)

Citizens and customers who need to report issues or complaints, and organizational administrators/agents who need to manage, assign, and resolve those complaints.

2. JOBS-TO-BE-DONE / PROBLEMS

Users need to securely register/log in, submit complaints with category and attachments, track status in real time, and give feedback after resolution. Admins need to view all complaints, assign agents, update statuses, and monitor resolution statistics.

3. TRIGGERS

Encountering a product, service, or civic issue with no clear channel to report it; on the admin side, complaints piling up with no central visibility into who is handling what.

4. EMOTIONS: BEFORE / AFTER

Before: Frustrated and ignored, unsure if the complaint is even being handled. After: Reassured by visible status tracking, satisfied once resolution is confirmed and feedback given.

5. AVAILABLE SOLUTIONS

Phone-based complaint desks, generic email support inboxes, physical complaint registers, and generic third-party ticketing tools not tailored to the organization's workflow.

6. CUSTOMER CONSTRAINTS

Requires a system baseline setup of Node.js, npm package manager dependencies, a running MongoDB instance (local or Atlas), and a configured .env with JWT secret and DB URI.

7. BEHAVIOUR

Asynchronously calling backend REST endpoints (auth, complaints, agents, feedback) using an Axios HTTP client, authenticated with a JWT bearer token stored client-side.

8. CHANNELS OF BEHAVIOUR

8.1 ONLINE: React web app (localhost:3000), Express REST API (localhost:5000).

8.2 OFFLINE: Local development setup via npm install / npm run dev, and seeding admin/agent accounts via npm run create-admin.

9. PROBLEM ROOT CAUSE

Complaint handling that relies on disconnected channels (calls, emails, walk-ins) with no shared status data, leaving no single, secure, role-based record of who is responsible for what.

10. SOLUTION

Online Complaint Registration and Management System: A MERN stack application (MongoDB, Express, React, Node.js) with JWT-based role authentication for Users, Agents, and Admins, a defined complaint status workflow (OPEN → IN_PROGRESS → RESOLVED → CLOSED), admin-driven agent assignment, and post-resolution feedback that automatically closes the complaint.